

Sliper Kod Dokumu

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Dosya sayisi: 20

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CMakeLists.txt

```
cmake_minimum_required(VERSION 3.16)
project(sliper VERSION 1.0)

set(CMAKE_CXX_STANDARD 17)
set(CMAKE_PREFIX_PATH "C:/Qt/6.7.3/mingw_64")

find_package(Qt6 REQUIRED COMPONENTS Core Quick Charts Widgets Sql PrintSupport)

qt_standard_project_setup()
qt_policy(SET QTP0001 NEW)

qt_add_executable(sliper
    main.cpp
    src/Backend.h
    src/Backend.cpp
    src/SensorManager.h
    src/SensorManager.cpp
    src/Calculator.h
    src/Calculator.cpp
    src/Database.h
    src/Database.cpp
    src/ReportManager.h
    src/ReportManager.cpp
)

qt_add_qml_module(sliper
    URI sliper
    VERSION 1.0

    QML_FILES
        qml/Main.qml
        qml/pages/LoginPage.qml
        qml/pages/DashboardPage.qml
        qml/pages/OlcumPage.qml
        qml/pages/SonuclarPage.qml
        qml/pages/GecmisPage.qml
        qml/pages/KalibrasyonPage.qml
```

```
    RESOURCES
        qml/assets/logo.png
        qml/assets/egim_sensor.png
        qml/assets/loadcell.png
        qml/assets/mesafe_sensor.png
    )

    target_link_libraries(sliper PRIVATE
        Qt6::Core
        Qt6::Quick
        Qt6::Charts
        Qt6::Widgets
        Qt6::Sql
        Qt6::PrintSupport
    )
```

esp32\sliper_esp32\sliper_esp32.ino

```

/*
 * SLIPER - ESP32-S3 Firmware (BLE UART Versiyonu)
 * -----
 * Sensorler:
 * - MPU6050      : Egim (X/Y derece)      -> I2C
 * - HX711 + S-tipi load cell : Basinc (mbar) -> Dijital (DT/SCK)
 * - ATEK LMS lazer : Mesafe (Analog 0-5V)  -> ADS1115 uzerinden I2C
 *
 * Cikis: Bluetooth Low Energy (BLE) uzerinden, her 200ms'de bir JSON paketi
 */

#include <Wire.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <HX711.h>
#include <Adafruit_ADS1X15.h>
#include <ArduinoJson.h>

// --- BLE Kutuphaneleri ---
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>

// ----- Pin Tanimlari -----
#define I2C_SDA 17
#define I2C_SCL 18

#define HX711_DT 39
#define HX711_SCK 40

// ----- Kalibrasyon Sabitleri -----
float hx711_kalibrasyonKatsayisi = 2280.0f;
long  hx711_sifirOffset = 0;

const float LAZER_MAX_MESAFE_MM = 2000.0f;
const float LAZER_MAX_VOLTAJ = 5.0f;

const float BORU_CAPI_M = 0.126f;
const float BORU_ALANI_M2 = 3.14159265f * (BORU_CAPI_M / 2.0f) * (BORU_CAPI_M / 2.0f);

// ----- Nesneler -----
Adafruit_MPU6050 mpu;
HX711 loadCell;
Adafruit_ADS1115 ads;

// ----- BLE Ayarlari -----
BLEServer *pServer = NULL;
BLECharacteristic *pTxCharacteristic;
bool deviceConnected = false;
bool oldDeviceConnected = false;

// Standart Nordic UART Service (NUS) UUID'leri
#define SERVICE_UUID          "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
#define CHARACTERISTIC_UUID_TX "6E400003-B5A3-F393-E0A9-E50E24DCCA9E"

class MyServerCallbacks: public BLEServerCallbacks {
    void onConnect(BLEServer* pServer) {
        deviceConnected = true;
    };

    void onDisconnect(BLEServer* pServer) {
        deviceConnected = false;
    }
};

// ----- Durum Degiskenleri -----
float onceki_konum_mm = 0.0f;
unsigned long onceki_zaman_ms = 0;

void setup() {
    Serial.begin(115200);
    Wire.begin(I2C_SDA, I2C_SCL);

```

```

// --- BLE Baslatma ---
BLEDevice::init("SLIPER-ESP32");
pServer = BLEDevice::createServer();
pServer->setCallbacks(new MyServerCallbacks());
BLEService *pService = pServer->createService(SERVICE_UUID);

// TX Karakteristigi (ESP32'den veri gondermek icin)
pTxCharacteristic = pService->createCharacteristic(
    CHARACTERISTIC_UUID_TX,
    BLECharacteristic::PROPERTY_NOTIFY
);
pTxCharacteristic->addDescriptor(new BLE2902());
pService->start();
pServer->getAdvertising()->start();
Serial.println("BLE baslatildi: SLIPER-ESP32 (Baglanti bekleniyor...)");

// MPU6050 baslat
if (!mpu.begin()) {
    Serial.println("MPU6050 bulunamadi!");
} else {
    mpu.setAccelerometerRange(MPU6050_RANGE_2_G);
    mpu.setGyroRange(MPU6050_RANGE_250_DEG);
    mpu.setFilterBandwidth(MPU6050_BAND_21_HZ);
    Serial.println("MPU6050 hazir.");
}

// HX711 baslat
loadCell.begin(HX711_DT, HX711_SCK);
if (loadCell.wait_ready_timeout(2000)) {
    hx711_sifirOffset = loadCell.read_average(10);
    Serial.println("HX711 hazir.");
} else {
    Serial.println("HX711 bulunamadi!");
}

// ADS1115 baslat
if (!ads.begin()) {
    Serial.println("ADS1115 bulunamadi!");
} else {
    ads.setGain(GAIN_ONE);
    Serial.println("ADS1115 hazir.");
}

onceki_zaman_ms = millis();
}

float basincOku() {
    if (!loadCell.is_ready()) return 0.0f;
    long ham = loadCell.read_average(3);
    float agirlikGram = (ham - hx711_sifirOffset) / hx711_kalibrasyonKatsayisi;
    return agirlikGram * 0.0981f;
}

float konumOku() {
    int16_t adcDeger = ads.readADC_SingleEnded(3);
    float voltaj = ads.computeVolts(adcDeger);
    if (voltaj < 0) voltaj = 0;
    if (voltaj > LAZER_MAX_VOLTAJ) voltaj = LAZER_MAX_VOLTAJ;
    return (voltaj / LAZER_MAX_VOLTAJ) * LAZER_MAX_MESAFE_MM;
}

void egimOku(float &egimX, float &egimY) {
    sensors_event_t ivme, gyro, sicaklik;
    mpu.getEvent(&ivme, &gyro, &sicaklik);
    egimX = atan2(ivme.acceleration.y, ivme.acceleration.z) * 180.0f / PI;
    egimY = atan2(-ivme.acceleration.x,
        sqrt(ivme.acceleration.y * ivme.acceleration.y +
            ivme.acceleration.z * ivme.acceleration.z)) * 180.0f / PI;
}

void loop() {
    float basinc = basincOku();
    float konum = konumOku();

```

```
unsigned long simdikiZaman = millis();
float dt = (simdikiZaman - onceki_zaman_ms) / 1000.0f;
if (dt <= 0) dt = 0.001f;

float hiz = fabs(konum - onceki_konum_mm) / 1000.0f / dt;
float debi = hiz * BORU_ALANI_M2 * 3600.0f;

onceki_konum_mm = konum;
onceki_zaman_ms = simdikiZaman;

float egimX = 0, egimY = 0;
egimOku(egimX, egimY);

StaticJsonDocument<256> doc;
doc["basinc"] = round(basinc * 100) / 100.0;
doc["konum"] = round(konum * 100) / 100.0;
doc["hiz"] = round(hiz * 100) / 100.0;
doc["debi"] = round(debi * 100) / 100.0;
doc["egimX"] = round(egimX * 100) / 100.0;
doc["egimY"] = round(egimY * 100) / 100.0;

String json;
serializeJson(doc, json);

// Sadece cihaz bagliyken BLE uzerinden veri gonder
if (deviceConnected) {
    pTxCharacteristic->setValue(json.c_str());
    pTxCharacteristic->notify(); // Veriyi karsi tarafa it (push)
}

Serial.println(json); // USB uzerinden hata ayiklama icin yazdir

// BLE baglantisi koptugunda tekrar gorunur olmasini sagla
if (!deviceConnected && oldDeviceConnected) {
    delay(500);
    pServer->startAdvertising();
    Serial.println("BLE cihazı koptu. Tekrar baglanti bekleniyor...");
    oldDeviceConnected = deviceConnected;
}
if (deviceConnected && !oldDeviceConnected) {
    oldDeviceConnected = deviceConnected;
}

delay(200);
}
```

main.cpp

```
#include <QmlContext>
#include <QApplication>
#include <QmlApplicationEngine>
#include <QDebug>
#include <QtGlobal>
#include "src/Backend.h"
#include "src/SensorManager.h"
#include "src/Calculator.h"
#include "src/Database.h"
#include "src/ReportManager.h"

int main(int argc, char *argv[])
{
    qputenv("QT_QUICK_CONTROLS_STYLE", "Basic");

    QApplication app(argc, argv);

    Backend backend;
    SensorManager sensorManager;
    Calculator calculator;
    Database database;
    ReportManager reportManager;

    QmlApplicationEngine engine;

    engine.rootContext()->setContextProperty("backend", &backend);
    engine.rootContext()->setContextProperty("sensorManager", &sensorManager);
    engine.rootContext()->setContextProperty("calculator", &calculator);
    engine.rootContext()->setContextProperty("database", &database);
    engine.rootContext()->setContextProperty("reportManager", &reportManager);

    QObject::connect(
        &engine,
        &QmlApplicationEngine::objectCreationFailed,
        &app,
        []() {
            qWarning() << "QML nesnesi olusturulamadi!";
            QCoreApplication::exit(-1);
        },
        Qt::QueuedConnection);

    engine.loadFromModule("sliper", "Main");

    return app.exec();
}
```

qml\Main.qml

```
import QtQuick 6.7
import QtQuick.Controls 6.7
import QtQuick.Window 6.7
import QtQuick.Layouts 6.7
import "pages"

ApplicationWindow {
    id: pencere
    visibility: Window.Maximized
    title: "SLIPER Analiz"

    background: Rectangle {
        color: "#060d17"
    }

    StackLayout {
        id: anaStack
        anchors.fill: parent
        currentIndex: 0

        LoginPage {
            onGirisBasarili: anaStack.currentIndex = 1
        }

        DashboardPage {
        }
    }
}
```

qml\pages\DashboardPage.qml

```

import QtQuick 6.7
import QtQuick.Controls 6.7
import QtQuick.Layouts 6.7

Rectangle {
    color: "#0a0e17"

    readonly property var navOgeleri: [
        { etiket: "A-lAŞm", ikon: "ğŸ“Š" },
        { etiket: "SonuAşlar", ikon: "ğŸ“^" },
        { etiket: "GeAşmiAş", ikon: "ğŸ•’" },
        { etiket: "Kalibrasyon", ikon: "âš“i,âš“" }
    ]

    Row {
        anchors.fill: parent
        spacing: 0

        Rectangle {
            id: sidebar
            width: 230
            height: parent.height

            gradient: Gradient {
                GradientStop { position: 0.0; color: "#121b30" }
                GradientStop { position: 1.0; color: "#080b12" }
            }

            Image {
                anchors.top: parent.top
                anchors.horizontalCenter: parent.horizontalCenter
                anchors.topMargin: 24
                source: "../assets/logo.png"
                width: 144
                height: 144
                fillMode: Image.PreserveAspectFit
            }

            Rectangle {
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.right: parent.right
                anchors.leftMargin: 24
                anchors.rightMargin: 24
                anchors.topMargin: 188
                height: 1
                color: "#1f2c46"
            }
        }

        Column {
            anchors.top: parent.top
            anchors.topMargin: 206
            width: parent.width
            spacing: 2

            Repeater {
                model: navOgeleri

                Rectangle {
                    id: navOgesi
                    width: parent.width
                    height: 46
                    property bool aktif: icerikStack.currentIndex === index
                    property bool hoverli: false
                    color: aktif ? "#182644" : (hoverli ? "#101a2c" : "transparent")
                    Behavior on color { ColorAnimation { duration: 120 } }

                    Rectangle {
                        width: 3
                        height: parent.height
                        color: "#3b82f6"
                        visible: navOgesi.aktif
                    }
                }
            }
        }
    }
}

```



```

        Row {
            anchors.left: parent.left
            anchors.leftMargin: 24
            anchors.verticalCenter: parent.verticalCenter
            spacing: 12

            Text {
                text: modelData.ikon
                font.pixelSize: 15
                anchors.verticalCenter: parent.verticalCenter
            }

            Text {
                text: modelData.etiket
                color: navOgesi.aktif ? "#6ea8ff" : (navOgesi.hoverli ? "#dce8f5" : "#9ca3af")
                font.family: "Segoe UI"
                font.pixelSize: 14
                font.bold: navOgesi.aktif
                anchors.verticalCenter: parent.verticalCenter
                Behavior on color { ColorAnimation { duration: 120 } }
            }
        }

        MouseArea {
            anchors.fill: parent
            hoverEnabled: true
            cursorShape: Qt.PointingHandCursor
            onEntered: navOgesi.hoverli = true
            onExited: navOgesi.hoverli = false
            onClicked: icerikStack.currentIndex = index
        }
    }
}

Rectangle {
    id: sidebarAyraci
    width: 3
    height: parent.height
    color: "#05070b"

    Rectangle {
        anchors.right: parent.right
        anchors.top: parent.top
        anchors.bottom: parent.bottom
        width: 1
        color: "#403b82f6"
    }
}

StackLayout {
    id: icerikStack
    width: parent.width - sidebar.width - sidebarAyraci.width
    height: parent.height
    currentIndex: 0

    OlcumPage {}

    SonuclarPage {
        id: sonuclarSayfasi
    }

    GecmisPage {
        onTestSecildi: function(id) {
            sonuclarSayfasi.olcumId = id
            icerikStack.currentIndex = 1
        }
    }

    KalibrasyonPage {}
}
}
}

```

qml\pages\GecmisPage.qml

```

import QtQuick 6.7
import QtQuick.Controls 6.7

Rectangle {
    color: "#0a0e17"

    signal testSecildi(int id)

    Column {
        anchors.fill: parent
        anchors.margins: 20
        spacing: 16

        Row {
            width: parent.width
            spacing: 12

            TextField {
                id: aramaKutusu
                width: parent.width - filtreKutusu.width - parent.spacing
                height: 40
                placeholderText: "MÃĖter i veya reÃŖete ara..."
                placeholderTextColor: "#4b5563"
                color: "ffffff"
                font.pixelSize: 13
                leftPadding: 34
                verticalAlignment: TextInput.AlignVCenter
                background: Rectangle {
                    color: "#0f1420"
                    radius: 8
                    border.color: aramaKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
                    border.width: 1
                }

                Text {
                    anchors.left: parent.left
                    anchors.leftMargin: 12
                    anchors.verticalCenter: parent.verticalCenter
                    text: "gŸ"
                    color: "#4b5563"
                    font.pixelSize: 13
                }
            }

            ComboBox {
                id: filtreKutusu
                width: 160
                height: 40
                model: ["TÃĖm Testler", "Son 7 GÃĖn", "Son 30 GÃĖn"]

                background: Rectangle {
                    color: "#0f1420"
                    radius: 8
                    border.color: filtreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
                    border.width: 1
                }

                contentItem: Text {
                    text: filtreKutusu.displayText
                    color: "#9ca3af"
                    font.pixelSize: 13
                    leftPadding: 10
                    verticalAlignment: Text.AlignVCenter
                }
            }
        }
    }

    ListView {
        width: parent.width
        height: parent.height - 56
        clip: true
        spacing: 10
    }
}

```

```

model: database.tumOlcumleriGetir()

delegate: Rectangle {
    id: gecmiskarti
    width: parent.width
    height: 76
    radius: 10
    color: "#0f1420"
    border.color: kartAlani.containsMouse ? "#3b82f6" : "#1e2a3f"
    border.width: 1
    Behavior on border.color { ColorAnimation { duration: 120 } }

    MouseArea {
        id: kartAlani
        anchors.fill: parent
        hoverEnabled: true
        acceptedButtons: Qt.NoButton
    }

    Rectangle {
        id: rozet
        width: 44
        height: 44
        radius: 10
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        anchors.leftMargin: 16
        color: "#132335"

        Text {
            anchors.centerIn: parent
            text: modelData.musteri.charAt(0).toUpperCase()
            color: "#3b82f6"
            font.family: "Segoe UI"
            font.pixelSize: 18
            font.bold: true
        }
    }
}

Column {
    anchors.left: rozet.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.leftMargin: 14
    spacing: 4

    Row {
        spacing: 8

        Text {
            text: modelData.tarih
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 11
        }

        Rectangle {
            width: strokeMetni.implicitWidth + 12
            height: 16
            radius: 8
            color: "#2a2010"
            anchors.verticalCenter: parent.verticalCenter

            Text {
                id: strokeMetni
                anchors.centerIn: parent
                text: modelData.strokeSayisi + " stroke"
                color: "#e8a020"
                font.family: "Segoe UI"
                font.pixelSize: 10
                font.bold: true
            }
        }
    }
}

Text {

```

```

        text: "MÃÄÿteri: " + modelData.musteri + " â€¢ ReÃ§ete: " + modelData.recete
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }
}

Button {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 16
    height: 34
    text: "GÃ¼rÃ¼ntÃ¼le"
    font.pixelSize: 12

    onClicked: testSecildi(modelData.id)

    background: Rectangle {
        radius: 6
        color: parent.hovered ? "#4f8cf7" : "#3b82f6"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
        leftPadding: 8
        rightPadding: 8
    }
}
}
}
}
```

qml\pages\KalibrasyonPage.qml

```

import QtQuick 6.7
import QtQuick.Controls 6.7
import QtQuick.Layouts 6.7

Rectangle {
    color: "#0a0e17"

    property int seciliSensor: -1
    property int egimAdimi: 0
    property int loadCellAdimi: 0
    property int mesafeAdimi: 0

    property bool egimKayitliVar: false
    property string egimSonTarih: ""

    property bool loadCellKayitliVar: false
    property real loadCellReferans: 0
    property string loadCellSonTarih: ""

    property bool mesafeUstKayitliVar: false
    property bool mesafeAltKayitliVar: false
    property string mesafeSonTarih: ""

    onSeciliSensorChanged: {
        if (seciliSensor === 0) {
            var e = database.kalibrasyonGetir("egim")
            egimKayitliVar = e.mevcut
            egimSonTarih = e.mevcut ? e.tarih : ""
        } else if (seciliSensor === 1) {
            var l = database.kalibrasyonGetir("loadcell")
            loadCellKayitliVar = l.mevcut
            loadCellReferans = l.mevcut ? l.deger1 : 0
            loadCellSonTarih = l.mevcut ? l.tarih : ""
        } else if (seciliSensor === 2) {
            var u = database.kalibrasyonGetir("mesafe_ust")
            var a = database.kalibrasyonGetir("mesafe_alt")
            mesafeUstKayitliVar = u.mevcut
            mesafeAltKayitliVar = a.mevcut
            mesafeSonTarih = u.mevcut ? u.tarih : (a.mevcut ? a.tarih : "")
        }
    }

    StackLayout {
        anchors.fill: parent
        currentIndex: seciliSensor === -1 ? 0 : 1

        Item {
            Text {
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.margins: 20
                text: "KALİBRASYON â€” Sensör SeâŸin"
                color: "#6b7280"
                font.family: "Segoe UI"
                font.pixelSize: 12
                font.bold: true
                font.letterSpacing: 1
            }

            Row {
                anchors.centerIn: parent
                spacing: 24

                Rectangle {
                    id: egimKarti
                    width: 350
                    height: 450
                    radius: 12
                    color: "#0f1420"
                    border.color: egimKarti.hovered ? "#14b8a6" : "#1e2a3f"
                    border.width: 1
                    property bool hovered: false
                    scale: hovered ? 1.03 : 1.0
                }
            }
        }
    }
}

```

```

Behavior on border.color { ColorAnimation { duration: 120 } }
Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }

Column {
    anchors.centerIn: parent
    spacing: 14

    Item {
        anchors.horizontalCenter: parent.horizontalCenter
        width: 210
        height: 150

        Image {
            anchors.centerIn: parent
            width: parent.width
            height: parent.height
            source: "../assets/egim_sensor.png"
            fillMode: Image.PreserveAspectFit
        }
    }

    Text { anchors.horizontalCenter: parent.horizontalCenter; text: "EÄÝim SensÃrÄ"; color: "#dce8f5";
font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
    Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Yatay hizalama kalibrasyonu"; color:
"#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        visible: egimKayitliVar
        width: kaliMetni1.implicitWidth + 16
        height: 20
        radius: 10
        color: "#123321"

        Text {
            id: kaliMetni1
            anchors.centerIn: parent
            text: "â€œ Kalibre Edildi"
            color: "#4ade80"
            font.pixelSize: 10
            font.bold: true
        }
    }
}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: egimKarti.hovered = true
    onExited: egimKarti.hovered = false
    onClicked: { seciliSensor = 0; egimAdimi = 0 }
}

Rectangle {
    id: loadCellKarti
    width: 350
    height: 450
    radius: 12
    color: "#0f1420"
    border.color: loadCellKarti.hovered ? "#3b82f6" : "#1e2a3f"
    border.width: 1
    property bool hovered: false
    scale: hovered ? 1.03 : 1.0
    Behavior on border.color { ColorAnimation { duration: 120 } }
    Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }

    Column {
        anchors.centerIn: parent
        spacing: 14

        Item {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 210

```

```

        height: 150

        Image {
            anchors.centerIn: parent
            width: parent.width
            height: parent.height
            source: "../assets/loadcell.png"
            fillMode: Image.PreserveAspectFit
        }
    }

    Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Load Cell"; color: "#dce8f5";
font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
    Text { anchors.horizontalCenter: parent.horizontalCenter; text: "BasÄ±nÄ± sensÃ¶rÃ¼ kalibrasyonu";
color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        visible: loadCellKayitliVar
        width: kaliMetni2.implicitWidth + 16
        height: 20
        radius: 10
        color: "#123321"

        Text {
            id: kaliMetni2
            anchors.centerIn: parent
            text: "â€œ Kalibre Edildi"
            color: "#4ade80"
            font.pixelSize: 10
            font.bold: true
        }
    }
}

MouseArea {
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onEntered: loadCellKarti.hovered = true
    onExited: loadCellKarti.hovered = false
    onClicked: { seciliSensor = 1; loadCellAdimi = 0 }
}

Rectangle {
    id: mesafeKarti
    width: 350
    height: 450
    radius: 12
    color: "#0f1420"
    border.color: mesafeKarti.hovered ? "#9333ea" : "#1e2a3f"
    border.width: 1
    property bool hovered: false
    scale: hovered ? 1.03 : 1.0
    Behavior on border.color { ColorAnimation { duration: 120 } }
    Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }

    Column {
        anchors.centerIn: parent
        spacing: 14

        Item {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 210
            height: 150

            Image {
                anchors.centerIn: parent
                width: parent.width
                height: parent.height
                source: "../assets/mesafe_sensor.png"
                fillMode: Image.PreserveAspectFit
            }
        }
    }
}

```

```

        Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Mesafe Sensör"; color: "#dce8f5";
font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
        Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Konum Kalibrasyonu";
color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            visible: mesafeUstKayitliVar && mesafeAltKayitliVar
            width: kaliMetni3.implicitWidth + 16
            height: 20
            radius: 10
            color: "#123321"

            Text {
                id: kaliMetni3
                anchors.centerIn: parent
                text: "Kalibre Edildi"
                color: "#4ade80"
                font.pixelSize: 10
                font.bold: true
            }
        }
    }

    MouseArea {
        anchors.fill: parent
        hoverEnabled: true
        cursorShape: Qt.PointingHandCursor
        onEntered: mesafeKarti.hovered = true
        onExited: mesafeKarti.hovered = false
        onClicked: { seciliSensor = 2; mesafeAdimi = 0 }
    }
}

Item {
    Row {
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.margins: 20
        spacing: 12

        Button {
            text: "Geri"
            font.pixelSize: 13

            background: Rectangle {
                radius: 6
                color: parent.hovered ? "#1e2a3f" : "#0f1420"
                border.color: "#1e2a3f"
                border.width: 1
                Behavior on color { ColorAnimation { duration: 120 } }
            }

            contentItem: Text {
                text: parent.text
                color: "#dce8f5"
                font: parent.font
                horizontalAlignment: Text.AlignHCenter
                verticalAlignment: Text.AlignVCenter
                leftPadding: 10
                rightPadding: 10
            }

            onClicked: seciliSensor = -1
        }

        Rectangle {
            width: 8
            height: 8
            radius: 4
            anchors.verticalCenter: parent.verticalCenter
            color: {

```



```

        if (seciliSensor === 0) return "#14b8a6"
        if (seciliSensor === 1) return "#3b82f6"
        if (seciliSensor === 2) return "#9333ea"
        return "#6b7280"
    }
}

Text {
    anchors.verticalCenter: parent.verticalCenter
    text: {
        if (seciliSensor === 0) return "EÄŸim Sensör Kalibrasyonu"
        if (seciliSensor === 1) return "Load Cell Kalibrasyonu"
        if (seciliSensor === 2) return "Mesafe Sensör Kalibrasyonu"
        return ""
    }
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 16
    font.bold: true
}

Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 70
    visible: seciliSensor === 0

    Column {
        anchors.centerIn: parent
        spacing: 20
        width: 360

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            visible: egimKayitliVar
            width: sonKaliMetni1.implicitWidth + 20
            height: 26
            radius: 13
            color: "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                id: sonKaliMetni1
                anchors.centerIn: parent
                text: "Son kalibrasyon: " + egimSonTarih
                color: "#6b7280"
                font.pixelSize: 11
            }
        }

        Text {
            width: parent.width
            horizontalAlignment: Text.AlignHCenter
            wrapMode: Text.WordWrap
            text: "Cihaz dâ€ž ve sabit bir yâ€žzeyeye yerleâ€žtirin"
            color: "#dce8f5"
            font.family: "Segoe UI"
            font.pixelSize: 14
        }

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 220
            height: 72
            radius: 12
            color: "#0f2e2a"
            border.color: "#14b8a6"
            border.width: 1

            Text {
                anchors.centerIn: parent

```

```

        text: "X: 0.0° Y: 0.0°"
        color: "#2dd4bf"
        font.family: "Segoe UI"
        font.pixelSize: 22
        font.bold: true
    }
}

Button {
    width: parent.width
    height: 44
    text: "Sıfırla (Zero)"
    font.pixelSize: 14
    font.bold: true

    onClicked: {
        database.kalibrasyonKaydet("egim", 0.0, 0.0)
        console.log("Egim sensoru kalibrasyonu kaydedildi.")
        seciliSensor = -1
    }

    background: Rectangle {
        radius: 8
        color: parent.hovered ? "#4f8cf7" : "#3b82f6"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "#ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

}

Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 70
    visible: seciliSensor === 1

    Column {
        anchors.centerIn: parent
        spacing: 20
        width: 360

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            visible: loadCellKayitliVar
            width: sonKaliMetni2.implicitWidth + 20
            height: 26
            radius: 13
            color: "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                id: sonKaliMetni2
                anchors.centerIn: parent
                text: "Son referans: " + loadCellReferans.toFixed(1) + " kg â€” " + loadCellSonTarih
                color: "#6b7280"
                font.pixelSize: 11
            }
        }

        Text {
            anchors.horizontalCenter: parent.horizontalCenter
            text: "Adım " + (loadCellAdimi + 1) + " / 2"
            color: "#6b7280"
            font.family: "Segoe UI"
        }
    }
}

```

```

        font.pixelSize: 11
        font.letterSpacing: 1
    }

    Row {
        anchors.horizontalCenter: parent.horizontalCenter
        spacing: 6

        Repeater {
            model: 2
            Rectangle {
                width: index === loadCellAdimi ? 22 : 8
                height: 8
                radius: 4
                color: index <= loadCellAdimi ? "#3b82f6" : "#1e2a3f"
                Behavior on width { NumberAnimation { duration: 150 } }
            }
        }
    }

    Column {
        width: parent.width
        spacing: 20
        visible: loadCellAdimi === 0

        Text {
            width: parent.width
            horizontalAlignment: Text.AlignHCenter
            wrapMode: Text.WordWrap
            text: "Load cell Ã¼zerinde hiÃ§bir aÃ§Ã¼rlÃ¼k olmadÃ¼Ã¼ndan emin olun"
            color: "#dce8f5"
            font.family: "Segoe UI"
            font.pixelSize: 14
        }

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            width: 220
            height: 72
            radius: 12
            color: "#132335"
            border.color: "#3b82f6"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: "0.0 mbar"
                color: "#3b82f6"
                font.family: "Segoe UI"
                font.pixelSize: 22
                font.bold: true
            }
        }
    }

    Button {
        width: parent.width
        height: 44
        text: "SÃ¼rlÃ¼ ve Devam Et"
        font.pixelSize: 14
        font.bold: true

        background: Rectangle {
            radius: 8
            color: parent.hovered ? "#4f8cf7" : "#3b82f6"
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "ffffff"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }

```

```

        onClicked: loadCellAdimi = 1
    }
}

Column {
    width: parent.width
    spacing: 20
    visible: loadCellAdimi === 1

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Bilinen bir referans aÄŸÄ±r1Ä±ÄŸÄ± yerleÄŸtirin"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 14
    }

    Row {
        anchors.horizontalCenter: parent.horizontalCenter
        spacing: 12

        Button {
            width: 100
            height: 40
            text: "1.6 kg"
            font.pixelSize: 13

            onClicked: {
                database.kalibrasyonKaydet("loadcell", 1.6, 0.0)
                console.log("Load cell referans 1.6kg kaydedildi.")
                seciliSensor = -1
            }

            background: Rectangle {
                radius: 8
                color: parent.hovered ? "#1e2a3f" : "#0a0e17"
                border.color: "#1e2a3f"
                border.width: 1
                Behavior on color { ColorAnimation { duration: 120 } }
            }

            contentItem: Text {
                text: parent.text
                color: "#dce8f5"
                font: parent.font
                horizontalAlignment: Text.AlignHCenter
                verticalAlignment: Text.AlignVCenter
            }
        }

        Button {
            width: 100
            height: 40
            text: "4.8 kg"
            font.pixelSize: 13

            onClicked: {
                database.kalibrasyonKaydet("loadcell", 4.8, 0.0)
                console.log("Load cell referans 4.8kg kaydedildi.")
                seciliSensor = -1
            }

            background: Rectangle {
                radius: 8
                color: parent.hovered ? "#1e2a3f" : "#0a0e17"
                border.color: "#1e2a3f"
                border.width: 1
                Behavior on color { ColorAnimation { duration: 120 } }
            }

            contentItem: Text {
                text: parent.text
                color: "#dce8f5"
            }
        }
    }
}

```

```

        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}
}
}
}
}

Item {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 70
    visible: seciliSensor === 2

    Column {
        anchors.centerIn: parent
        spacing: 20
        width: 360

        Rectangle {
            anchors.horizontalCenter: parent.horizontalCenter
            visible: mesafeUstKayitliVar || mesafeAltKayitliVar
            width: sonKaliMetni3.implicitWidth + 20
            height: 26
            radius: 13
            color: "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                id: sonKaliMetni3
                anchors.centerIn: parent
                text: "Son kalibrasyon: " + mesafeSonTarih
                color: "#6b7280"
                font.pixelSize: 11
            }
        }

        Text {
            anchors.horizontalCenter: parent.horizontalCenter
            text: "AdÄ±m " + (mesafeAdimi + 1) + " / 2"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 11
            font.letterSpacing: 1
        }
    }

    Row {
        anchors.horizontalCenter: parent.horizontalCenter
        spacing: 6

        Repeater {
            model: 2
            Rectangle {
                width: index === mesafeAdimi ? 22 : 8
                height: 8
                radius: 4
                color: index <= mesafeAdimi ? "#9333ea" : "#1e2a3f"
                Behavior on width { NumberAnimation { duration: 150 } }
            }
        }
    }

    Column {
        width: parent.width
        spacing: 20
        visible: mesafeAdimi === 0

        Text {
            width: parent.width
            horizontalAlignment: Text.AlignHCenter

```

```

        wrapMode: Text.WordWrap
        text: "Boruyu Ã¼st pozisyona getirin ve sabitleyin"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 14
    }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        width: 220
        height: 72
        radius: 12
        color: "#241a38"
        border.color: "#9333ea"
        border.width: 1

        Text {
            anchors.centerIn: parent
            text: "0.0 mm"
            color: "#c084fc"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }
    }

    Button {
        width: parent.width
        height: 44
        text: "Ã¼st Pozisyonu Kaydet"
        font.pixelSize: 14
        font.bold: true

        onClicked: {
            database.kalibrasyonKaydet("mesafe_ust", 0.0, 0.0)
            mesafeAdimi = 1
        }

        background: Rectangle {
            radius: 8
            color: parent.hovered ? "#4f8cf7" : "#3b82f6"
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "#ffffff"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}

Column {
    width: parent.width
    spacing: 20
    visible: mesafeAdimi === 1

    Text {
        width: parent.width
        horizontalAlignment: Text.AlignHCenter
        wrapMode: Text.WordWrap
        text: "Boruyu alt pozisyona getirin ve sabitleyin"
        color: "#dce8f5"
        font.family: "Segoe UI"
        font.pixelSize: 14
    }

    Rectangle {
        anchors.horizontalCenter: parent.horizontalCenter
        width: 220
        height: 72
        radius: 12
        color: "#241a38"
    }
}

```


qml\pages\LoginPage.qml

```
import QtQuick 6.7
import QtQuick.Controls 6.7

Rectangle {
    id: sayfa

    property bool sifreGorunur: false
    property bool beniHatirla: false

    signal girisBasarili()

    gradient: Gradient {
        GradientStop { position: 0.0; color: "#101c33" }
        GradientStop { position: 1.0; color: "#0a1220" }
    }

    Row {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.margins: 24
        spacing: 8

        Rectangle {
            width: 44
            height: 30
            radius: 6
            color: "#0f1420"
            border.color: "#3b82f6"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: "TR"
                color: "ffffff"
                font.pixelSize: 12
                font.bold: true
            }
        }

        Rectangle {
            width: 44
            height: 30
            radius: 6
            color: "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1

            Text {
                anchors.centerIn: parent
                text: "EN"
                color: "#6b7280"
                font.pixelSize: 12
            }
        }
    }

    Rectangle {
        anchors.centerIn: kart
        width: kart.width + 48
        height: kart.height + 48
        radius: 28
        color: "#18000000"
    }

    Rectangle {
        anchors.centerIn: kart
        width: kart.width + 32
        height: kart.height + 32
        radius: 24
        color: "#25000000"
    }

    Rectangle {
```



```

        anchors.centerIn: kart
        width: kart.width + 18
        height: kart.height + 18
        radius: 20
        color: "#35000000"
    }

    Rectangle {
        anchors.centerIn: kart
        width: kart.width + 8
        height: kart.height + 8
        radius: 18
        color: "#45000000"
    }

    Rectangle {
        id: kart
        anchors.centerIn: parent
        width: 400
        color: "#0f1420"
        radius: 16
        border.color: "#4b5563"
        border.width: 1
        height: govde.implicitHeight + 48

        Column {
            id: govde
            anchors.top: parent.top
            anchors.horizontalCenter: parent.horizontalCenter
            anchors.topMargin: 32
            width: parent.width - 48
            spacing: 20

            Image {
                anchors.horizontalCenter: parent.horizontalCenter
                source: "../assets/logo.png"
                width: 144
                height: 144
                fillMode: Image.PreserveAspectFit
            }

            Rectangle {
                width: parent.width
                height: 1
                color: "#1e2a3f"
            }

            Column {
                width: parent.width
                spacing: 4

                Text {
                    anchors.horizontalCenter: parent.horizontalCenter
                    text: "Sisteme GiriÅ"
                    color: "#ffffff"
                    font.family: "Segoe UI"
                    font.pixelSize: 22
                    font.bold: true
                }

                Text {
                    anchors.horizontalCenter: parent.horizontalCenter
                    text: "Yetkili hesabÅnÅzla devam edin"
                    color: "#6b7280"
                    font.family: "Segoe UI"
                    font.pixelSize: 12
                }
            }
        }

        Column {
            width: parent.width
            spacing: 6

            Text {
                text: "KULLANICI ADI"
            }
        }
    }

```

```

        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    TextField {
        id: kullaniciAdiKutusu
        width: parent.width
        height: 42
        placeholderText: "kullanici_adi"
        placeholderTextColor: "#4b5563"
        color: "ffffff"
        font.pixelSize: 13
        leftPadding: 12
        verticalAlignment: TextInput.AlignVCenter
        background: Rectangle {
            color: "#0a0e17"
            radius: 6
            border.color: kullaniciAdiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
            border.width: 1
        }
    }
}

Column {
    width: parent.width
    spacing: 6

    Text {
        text: "Å°FRE"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    TextField {
        id: sifreKutusu
        width: parent.width
        height: 42
        placeholderText: "*****"
        placeholderTextColor: "#4b5563"
        echoMode: sayfa.sifreGorunur ? TextInput.Normal : TextInput.Password
        color: "ffffff"
        font.pixelSize: 13
        leftPadding: 12
        rightPadding: 48
        verticalAlignment: TextInput.AlignVCenter
        background: Rectangle {
            color: "#0a0e17"
            radius: 6
            border.color: sifreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
            border.width: 1
        }
    }

    Rectangle {
        id: sifreGosterButonu
        anchors.right: parent.right
        anchors.verticalCenter: parent.verticalCenter
        anchors.rightMargin: 8
        width: 28
        height: 28
        radius: 6
        color: sifreGosterMouse.pressed ? "#162033" : (sifreGosterMouse.containsMouse ? "#111827" :
"transparent")

        border.color: sayfa.sifreGorunur ? "#334155" : "transparent"
        border.width: 1

        property color ikonRengi: sifreGosterMouse.containsMouse || sayfa.sifreGorunur ? "#cbd5e1" :
"#6b7280"

        Canvas {
            id: sifreIkonu
            anchors.centerIn: parent

```

```

width: 18
height: 18

onPaint: {
    var ctx = getContext("2d")
    ctx.clearRect(0, 0, width, height)
    ctx.lineWidth = 1.7
    ctx.lineCap = "round"
    ctx.lineJoin = "round"
    ctx.strokeStyle = sifreGosterButonu.ikonRengi

    ctx.beginPath()
    ctx.moveTo(1.5, 9)
    ctx.bezierCurveTo(4.2, 4.8, 7.0, 3.8, 9, 3.8)
    ctx.bezierCurveTo(11.0, 3.8, 13.8, 4.8, 16.5, 9)
    ctx.bezierCurveTo(13.8, 13.2, 11.0, 14.2, 9, 14.2)
    ctx.bezierCurveTo(7.0, 14.2, 4.2, 13.2, 1.5, 9)
    ctx.stroke()

    ctx.beginPath()
    ctx.arc(9, 9, 2.5, 0, Math.PI * 2, false)
    ctx.stroke()

    if (sayfa.sifreGorunur) {
        ctx.beginPath()
        ctx.moveTo(3, 15)
        ctx.lineTo(15, 3)
        ctx.stroke()
    }
}

Connections {
    target: sayfa
    function onSifreGorunurChanged() {
        sifreIkonu.requestPaint()
    }
}

Connections {
    target: sifreGosterButonu
    function onIkonRengiChanged() {
        sifreIkonu.requestPaint()
    }
}

}

MouseArea {
    id: sifreGosterMouse
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
}

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 12
    text: sayfa.sifreGorunur ? "göster" : "gizle"
    color: "#6b7280"
    font.pixelSize: 14
    visible: false

    MouseArea {
        anchors.fill: parent
        anchors.margins: -8
        cursorShape: Qt.PointingHandCursor
        onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
    }
}
}

Button {

```

```

        width: parent.width
        height: 44
        text: "GiriÅŸ Yap"
        font.family: "Segoe UI"
        font.pixelSize: 14
        font.bold: true

        onClicked: {
            var sonuc = backend.girisYap(kullaniciAdiKutusu.text, sifreKutusu.text)

            if (sonuc) {
                sayfa.girisBasarili()
            } else {
                console.log("Kullanici adi veya sifre hatali")
            }
        }

        background: Rectangle {
            radius: 8
            color: parent.pressed ? "#2563eb" : (parent.hovered ? "#4f8cf7" : "#3b82f6")
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "#ffffff"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }

    Rectangle {
        width: parent.width
        height: 1
        color: "#1e2a3f"
    }

    Text {
        anchors.horizontalCenter: parent.horizontalCenter
        text: "\u00A9 2026 Liya Test A.ÅŸ. SLIPER Analiz YazÄ±lÄ±mÄ±"
        color: "#4b5563"
        font.family: "Segoe UI"
        font.pixelSize: 10
    }
}
}
}
}

```

qml\pages\OlcumPage.qml

```

import QtQuick 6.7
import QtQuick.Controls 6.7
import QtCharts 6.7

Rectangle {
    color: "#0a0e17"
    property real zamanSayaci: 0
    property int aktifOlcumId: -1

    Row {
        anchors.fill: parent
        spacing: 0

        Rectangle {
            id: testBilgileriPaneli
            width: 280
            height: parent.height
            color: "#0f1420"

            Text {
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.margins: 20
                text: "TEST BÄ°LGÄ°LERÄ°"
                color: "#6b7280"
                font.family: "Segoe UI"
                font.pixelSize: 12
                font.bold: true
                font.letterSpacing: 1
            }

            Column {
                id: formAlanlari
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.right: parent.right
                anchors.margins: 20
                anchors.topMargin: 60
                spacing: 6

                Text {
                    text: "MÄÄteri"
                    color: "#9ca3af"
                    font.family: "Segoe UI"
                    font.pixelSize: 12
                }

                TextField {
                    id: musteriKutusu
                    width: parent.width
                    height: 38
                    placeholderText: "Musteri adi girin..."
                    placeholderTextColor: "#4b5563"
                    color: "#ffffff"
                    font.pixelSize: 13
                    leftPadding: 10
                    verticalAlignment: TextInput.AlignVCenter
                    background: Rectangle {
                        color: "#0a0e17"
                        radius: 6
                        border.color: musteriKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
                        border.width: 1
                    }
                }

                Text {
                    text: "Beton ReÄsetesi"
                    color: "#9ca3af"
                    font.family: "Segoe UI"
                    font.pixelSize: 12
                }

                ComboBox {

```

```

id: receteKutusu
width: parent.width
height: 38
model: ["ReÅŸete SeÅŸin...", "C25/30 Standart", "C30/37 SCC", "C35/45 Yuksek Mukavemet"]

background: Rectangle {
    color: "#0a0e17"
    radius: 6
    border.color: receteKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
    border.width: 1
}

contentItem: Text {
    text: receteKutusu.displayText
    color: "#9ca3af"
    font.pixelSize: 13
    leftPadding: 10
    verticalAlignment: Text.AlignVCenter
}
}

Text {
    text: "Eklenen AÄŸrlÄ±k (kg)"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 12
}

TextField {
    id: agirlikKutusu
    width: parent.width
    height: 38
    placeholderText: "orn: 4.8"
    placeholderTextColor: "#4b5563"
    color: "#ffffff"
    font.pixelSize: 13
    leftPadding: 10
    verticalAlignment: TextInput.AlignVCenter
    validator: DoubleValidator { bottom: 0; top: 20; decimals: 1 }
    background: Rectangle {
        color: "#0a0e17"
        radius: 6
        border.color: agirlikKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
        border.width: 1
    }
}

Item { width: parent.width; height: 10 }

Button {
    id: baslatButonu
    width: parent.width
    height: 44
    text: "Ä±lÄ±mlÄ± BaÅŸlat"
    font.family: "Segoe UI"
    font.pixelSize: 14
    font.bold: true

    onClicked: {
        if (!sensorManager.veriGecerli) {
            console.warn("Gercek sensor verisi yok, olcum baslatilmadi.")
            return
        }

        calculator.sifirla()
        aktifOlcumId = database.olcumBaslat(
            musteriKutusu.text,
            receteKutusu.currentText,
            parseFloat(agirlikKutusu.text)
        )
        console.log("Aktif olcum id:", aktifOlcumId)
    }

    background: Rectangle {
        radius: 8
    }
}

```

```

        color: baslatButonu.pressed ? "#15803d" : (baslatButonu.hovered ? "#22c55e" : "#16a34a")
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: baslatButonu.text
        color: "#ffffff"
        font: baslatButonu.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Row {
    width: parent.width
    spacing: 10

    Button {
        id: duraklatButonu
        width: (parent.width - parent.spacing) / 2
        height: 40
        text: calculator.duraklatildi ? "â–¶ Devam Et" : "â–¶ Duraklat"
        font.family: "Segoe UI"
        font.pixelSize: 13

        onClicked: {
            if (calculator.duraklatildi) {
                calculator.devamEt()
            } else {
                calculator.duraklat()
            }
        }

        background: Rectangle {
            radius: 8
            color: duraklatButonu.pressed ? "#78350f" : (duraklatButonu.hovered ? "#a16207" : "#92400e")
            border.color: "#d97706"
            border.width: 1
        }

        contentItem: Text {
            text: duraklatButonu.text
            color: "#ffffff"
            font: duraklatButonu.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }

    Button {
        id: bitirButonu
        width: (parent.width - parent.spacing) / 2
        height: 40
        text: "â–¶ Bitir"
        font.family: "Segoe UI"
        font.pixelSize: 13

        onClicked: {
            aktifOlcumId = -1
            console.log("Olcum sonlandirildi.")
        }

        background: Rectangle {
            radius: 8
            color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" : "#991b1b")
            border.color: "#dc2626"
            border.width: 1
        }

        contentItem: Text {
            text: bitirButonu.text
            color: "#ffffff"
            font: bitirButonu.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}

```

```

    }
  }
}

Rectangle {
    id: durumKarti
    anchors.top: formAlanlari.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.topMargin: 20
    anchors.leftMargin: 20
    anchors.rightMargin: 20
    height: 52
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1

    Row {
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        anchors.leftMargin: 14
        spacing: 10

        Rectangle {
            id: trafikIsigi
            width: 12
            height: 12
            radius: 6
            anchors.verticalCenter: parent.verticalCenter
            color: {
                if (!sensorManager.veriGecerli) return "#dc2626"
                if (calculator.duraklatildi) return "#6b7280"
                if (calculator.durum === "YUKARIDA") return "#16a34a"
                if (calculator.durum === "INIYOR") return "#f59e0b"
                return "#dc2626"
            }
        }

        Rectangle {
            anchors.centerIn: parent
            width: parent.width + 8
            height: parent.height + 8
            radius: width / 2
            color: "transparent"
            border.color: parent.color
            border.width: 1
            opacity: 0.35
        }
    }
}

Text {
    anchors.verticalCenter: parent.verticalCenter
    text: {
        if (!sensorManager.veriGecerli) return "VERI YOK"
        if (calculator.duraklatildi) return "DURAKLATILDI"
        return calculator.durum
    }
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}

Row {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 14
    spacing: 6

    Text {
        anchors.verticalCenter: parent.verticalCenter
        text: "STROKE"
        color: "#6b7280"
    }
}

```



```

        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    Text {
        anchors.verticalCenter: parent.verticalCenter
        text: calculator.strokeSayisi
        color: "#e8a020"
        font.family: "Segoe UI"
        font.pixelSize: 16
        font.bold: true
    }
}

Text {
    id: anlikDegerlerBaslik
    anchors.top: durumKarti.bottom
    anchors.left: parent.left
    anchors.topMargin: 24
    anchors.leftMargin: 20
    text: "ANLIK DEĞERLER"
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 12
    font.bold: true
    font.letterSpacing: 1
}

Grid {
    anchors.top: anlikDegerlerBaslik.bottom
    anchors.left: parent.left
    anchors.topMargin: 12
    anchors.leftMargin: 20
    columns: 2
    spacing: 10

    Rectangle {
        id: basincKarti
        width: 115
        height: 90
        radius: 10
        color: "#0a0e17"
        border.color: "#1e2a3f"
        border.width: 1
        clip: true

        Rectangle { width: 4; height: parent.height; color: "#3b82f6" }

        Column {
            anchors.top: parent.top
            anchors.right: parent.right
            anchors.bottom: parent.bottom
            anchors.left: parent.left
            anchors.topMargin: 10
            anchors.rightMargin: 10
            anchors.bottomMargin: 10
            anchors.leftMargin: 16
            spacing: 6

            Text {
                text: "BASINÇ"
                color: "#6b7280"
                font.family: "Segoe UI"
                font.pixelSize: 10
                font.letterSpacing: 1
            }

            Text {
                text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) : "--"
                color: "#3b82f6"
                font.family: "Segoe UI"
                font.pixelSize: 22
                font.bold: true
            }
        }
    }
}

```

```

    }

    Text {
        text: "mbar"
        color: "#4b5563"
        font.family: "Segoe UI"
        font.pixelSize: 10
    }
}

Rectangle {
    width: 115
    height: 90
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#9333ea" }

    Column {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.left: parent.left
        anchors.topMargin: 10
        anchors.rightMargin: 10
        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "KONUM"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) : "--"
            color: "#9333ea"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            text: "mm"
            color: "#4b5563"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

Rectangle {
    width: 115
    height: 90
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#f59e0b" }

    Column {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.left: parent.left
        anchors.topMargin: 10
    }
}

```

```

        anchors.rightMargin: 10
        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "HIZ"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) : "--"
            color: "#f59e0b"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            text: "m/s"
            color: "#4b5563"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

Rectangle {
    width: 115
    height: 90
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#16a34a" }

    Column {
        anchors.top: parent.top
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.left: parent.left
        anchors.topMargin: 10
        anchors.rightMargin: 10
        anchors.bottomMargin: 10
        anchors.leftMargin: 16
        spacing: 6

        Text {
            text: "DEBÄ°"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) : "--"
            color: "#16a34a"
            font.family: "Segoe UI"
            font.pixelSize: 22
            font.bold: true
        }

        Text {
            text: "m3/h"
            color: "#4b5563"
            font.family: "Segoe UI"
            font.pixelSize: 10
        }
    }
}

```

```

    }
  }
}

Rectangle {
    id: panelAyraci
    width: 3
    height: parent.height
    color: "#05070b"

    Rectangle {
        anchors.right: parent.right
        anchors.top: parent.top
        anchors.bottom: parent.bottom
        width: 1
        color: "#403b82f6"
    }
}

Item {
    width: parent.width - testBilgileriPaneli.width - panelAyraci.width
    height: parent.height

    Grid {
        anchors.fill: parent
        anchors.margins: 16
        columns: 2
        spacing: 16

        Rectangle {
            width: (parent.width - parent.spacing) / 2
            height: (parent.height - parent.spacing) / 2
            radius: 10
            color: "#0f1420"
            border.color: "#17263d"
            border.width: 1

            Item {
                id: basincBaslik
                anchors.top: parent.top
                anchors.left: parent.left
                anchors.right: parent.right
                anchors.margins: 14
                height: 20

                Rectangle {
                    id: basincNoktasi
                    width: 8
                    height: 8
                    radius: 4
                    color: "#3b82f6"
                    anchors.left: parent.left
                    anchors.verticalCenter: parent.verticalCenter
                }

                Text {
                    anchors.left: basincNoktasi.right
                    anchors.leftMargin: 8
                    anchors.verticalCenter: parent.verticalCenter
                    text: "Basıncın Zaman"
                    color: "#9ca3af"
                    font.family: "Segoe UI"
                    font.pixelSize: 13
                    font.bold: true
                }

                Text {
                    anchors.right: parent.right
                    anchors.verticalCenter: parent.verticalCenter
                    text: sensorManager.verigecerli ? sensorManager.basinc.toFixed(1) + " mbar" : "-- mbar"
                    color: "#3b82f6"
                    font.family: "Segoe UI"
                    font.pixelSize: 13
                    font.bold: true
                }
            }
        }
    }
}

```

```

    }

    ChartView {
        anchors.top: basincBaslik.bottom
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.topMargin: 4
        anchors.bottomMargin: 4
        backgroundColor: "transparent"
        legend.visible: false
        antialiasing: true

        ValueAxis {
            id: xEkseni
            min: 0
            max: 60
            gridLineColor: "#182131"
            labelsColor: "#4b5563"
            labelsFont.pixelSize: 9
            lineVisible: false
            minorGridVisible: false
        }

        ValueAxis {
            id: yEkseni
            min: 0
            max: 30
            gridLineColor: "#182131"
            labelsColor: "#4b5563"
            labelsFont.pixelSize: 9
            lineVisible: false
            minorGridVisible: false
        }

        SplineSeries {
            id: basincSerisi
            axisX: xEkseni
            axisY: yEkseni
            color: "#3b82f6"
            width: 2
        }

        Connections {
            target: sensorManager
            function onBasincChanged() {
                if (!sensorManager.veriGecerli) return

                zamanSayaci += 0.2
                basincSerisi.append(zamanSayaci, sensorManager.basinc)

                if (zamanSayaci > 60) {
                    basincSerisi.remove(0)
                    xEkseni.min = zamanSayaci - 60
                    xEkseni.max = zamanSayaci
                }
            }
        }
    }

    Rectangle {
        width: (parent.width - parent.spacing) / 2
        height: (parent.height - parent.spacing) / 2
        radius: 10
        color: "#0f1420"
        border.color: "#241a38"
        border.width: 1

        Item {
            id: konumBaslik
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.right: parent.right
            anchors.margins: 14

```

```
height: 20

Rectangle {
    id: konumNoktasi
    width: 8
    height: 8
    radius: 4
    color: "#9333ea"
    anchors.left: parent.left
    anchors.verticalCenter: parent.verticalCenter
}

Text {
    anchors.left: konumNoktasi.right
    anchors.leftMargin: 8
    anchors.verticalCenter: parent.verticalCenter
    text: "Konum - Zaman"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) + " mm" : "-- mm"
    color: "#9333ea"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}
}

ChartView {
    anchors.top: konumBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.bottomMargin: 4
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: konumXEkseni
        min: 0
        max: 60
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ValueAxis {
        id: konumYEkseni
        min: 0
        max: 500
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    SplineSeries {
        id: konumSerisi
        axisX: konumXEkseni
        axisY: konumYEkseni
        color: "#9333ea"
        width: 2
    }
}
```

```

Connections {
    target: sensorManager
    function onKonumChanged() {
        if (!sensorManager.veriGecerli) return

        konumSerisi.append(zamanSayaci, sensorManager.konum)

        if (zamanSayaci > 60) {
            konumSerisi.remove(0)
            konumXEkseni.min = zamanSayaci - 60
            konumXEkseni.max = zamanSayaci
        }

        calculator.konumGuncelle(sensorManager.konum)
    }
}

Connections {
    target: calculator
    function onStrokeSayisiChanged() {
        if (aktifOlcumId > 0 && sensorManager.veriGecerli) {
            database.strokeKaydet(aktifOlcumId, sensorManager.basinc, sensorManager.konum,
sensorManager.debi, true)
        }
    }
}

Rectangle {
    width: (parent.width - parent.spacing) / 2
    height: (parent.height - parent.spacing) / 2
    radius: 10
    color: "#0f1420"
    border.color: "#3a2a14"
    border.width: 1

    Item {
        id: hizBaslik
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.margins: 14
        height: 20

        Rectangle {
            id: hizNoktasi
            width: 8
            height: 8
            radius: 4
            color: "#f59e0b"
            anchors.left: parent.left
            anchors.verticalCenter: parent.verticalCenter
        }

        Text {
            anchors.left: hizNoktasi.right
            anchors.leftMargin: 8
            anchors.verticalCenter: parent.verticalCenter
            text: "Hız - Zaman"
            color: "#9ca3af"
            font.family: "Segoe UI"
            font.pixelSize: 13
            font.bold: true
        }

        Text {
            anchors.right: parent.right
            anchors.verticalCenter: parent.verticalCenter
            text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) + " m/s" : "-- m/s"
            color: "#f59e0b"
            font.family: "Segoe UI"
            font.pixelSize: 13
            font.bold: true
        }
    }
}

```

```

    }

    ChartView {
        anchors.top: hizBaslik.bottom
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.topMargin: 4
        anchors.bottomMargin: 4
        backgroundColor: "transparent"
        legend.visible: false
        antialiasing: true

        ValueAxis {
            id: hizXEkseni
            min: 0
            max: 60
            gridLineColor: "#182131"
            labelsColor: "#4b5563"
            labelsFont.pixelSize: 9
            lineVisible: false
            minorGridVisible: false
        }

        ValueAxis {
            id: hizYEkseni
            min: 0
            max: 4
            gridLineColor: "#182131"
            labelsColor: "#4b5563"
            labelsFont.pixelSize: 9
            lineVisible: false
            minorGridVisible: false
        }

        SplineSeries {
            id: hizSerisi
            axisX: hizXEkseni
            axisY: hizYEkseni
            color: "#f59e0b"
            width: 2
        }

        Connections {
            target: sensorManager
            function onHizChanged() {
                if (!sensorManager.veriGecerli) return

                hizSerisi.append(zamanSayaci, sensorManager.hiz)

                if (zamanSayaci > 60) {
                    hizSerisi.remove(0)
                    hizXEkseni.min = zamanSayaci - 60
                    hizXEkseni.max = zamanSayaci
                }
            }
        }
    }
}

Rectangle {
    width: (parent.width - parent.spacing) / 2
    height: (parent.height - parent.spacing) / 2
    radius: 10
    color: "#0f1420"
    border.color: "#163321"
    border.width: 1

    Item {
        id: debiBaslik
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.margins: 14
        height: 20
    }
}

```



```

Rectangle {
    id: debiNoktasi
    width: 8
    height: 8
    radius: 4
    color: "#16a34a"
    anchors.left: parent.left
    anchors.verticalCenter: parent.verticalCenter
}

Text {
    anchors.left: debiNoktasi.right
    anchors.leftMargin: 8
    anchors.verticalCenter: parent.verticalCenter
    text: "Debi - Zaman"
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) + " m3/h" : "-- m3/h"
    color: "#16a34a"
    font.family: "Segoe UI"
    font.pixelSize: 13
    font.bold: true
}
}

ChartView {
    anchors.top: debiBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.bottomMargin: 4
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: debiXEkseni
        min: 0
        max: 60
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ValueAxis {
        id: debiYEkseni
        min: 0
        max: 15
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    SplineSeries {
        id: debiSerisi
        axisX: debiXEkseni
        axisY: debiYEkseni
        color: "#16a34a"
        width: 2
    }

    Connections {

```

```
target: sensorManager
function onDebiChanged() {
    if (!sensorManager.veriGecerli) return

    debiSerisi.append(zamanSayaci, sensorManager.debi)

    if (zamanSayaci > 60) {
        debiSerisi.remove(0)
        debiXEkseni.min = zamanSayaci - 60
        debiXEkseni.max = zamanSayaci
    }
}
}
}
}
}
}
}
```

qml\pages\SonucIarPage.qml

```

import QtQuick 6.7
import QtQuick.Controls 6.7
import QtCharts 6.7

Rectangle {
    color: "#0a0e17"

    property int olcumId: -1
    property real hesaplananTau0: 0
    property real hesaplananMu: 0
    property int playbackAdimi: -1
    property bool oynatiliyor: false
    property int playbackHizi: 1

    onOlcumIdChanged: verileriYukle()

    function verileriYukle() {
        if (olcumId <= 0) {
            return
        }

        playbackAdimi = -1
        oynatiliyor = false
        playbackTimer.stop()

        var sonuc = database.binghamHesapla(olcumId)

        hesaplananTau0 = sonuc.tau0
        hesaplananMu = sonuc.mu

        tau0Metni.text = sonuc.tau0.toFixed(2) + " mbar"
        muMetni.text = sonuc.mu.toFixed(2) + " mbar·h/mÅ³"
        r2Metni.text = sonuc.r2.toFixed(2)

        durumKutusu.durumIyiMi = sonuc.yeterliVeri && sonuc.tau0 < 5 && sonuc.mu < 10
        durumKutusu.tau0Yuksek = sonuc.tau0 >= 5
        durumKutusu.muYuksek = sonuc.mu >= 10

        pqSerisi.clear()
        regresyonCizgisi.clear()
        oynatmaSerisi.clear()
        strokeModeli.clear()

        var strokeListesi = database.strokeVerileriGetir(olcumId)
        for (var i = 0; i < strokeListesi.length; i++) {
            var s = strokeListesi[i]
            pqSerisi.append(s.debi, s.basinc)
            strokeModeli.append({ stroke: s.stroke, p: s.basinc, q: s.debi, gecerli: s.gecerli })
        }

        regresyonCizgisi.append(0, sonuc.tau0)
        regresyonCizgisi.append(20, sonuc.tau0 + sonuc.mu * 20)
    }

    Timer {
        id: playbackTimer
        interval: 1000 / playbackHizi
        repeat: true
        running: false

        onTriggered: {
            if (playbackAdimi < strokeModeli.count - 1) {
                playbackAdimi++
                var satir = strokeModeli.get(playbackAdimi)
                oynatmaSerisi.clear()
                oynatmaSerisi.append(satir.q, satir.p)
            } else {
                oynatiliyor = false
                playbackTimer.stop()
            }
        }
    }
}

```

```

Row {
    anchors.fill: parent
    spacing: 0

    Rectangle {
        id: ozetPaneli
        width: 280
        height: parent.height
        color: "#0f1420"

        Text {
            anchors.top: parent.top
            anchors.left: parent.left
            anchors.margins: 20
            text: "SONUÂ# Ã-ZETÃ°"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 12
            font.bold: true
            font.letterSpacing: 1
        }
    }

    Flickable {
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.margins: 20
        anchors.topMargin: 60
        contentWidth: width
        contentHeight: sonucKartlari.height
        clip: true

        Column {
            id: sonucKartlari
            width: parent.width
            spacing: 10

            Rectangle {
                width: parent.width
                height: 70
                radius: 10
                color: "#0a0e17"
                border.color: "#17263d"
                border.width: 1
                clip: true

                Rectangle { width: 4; height: parent.height; color: "#3b82f6" }

                Column {
                    anchors.left: parent.left
                    anchors.verticalCenter: parent.verticalCenter
                    anchors.leftMargin: 20
                    spacing: 2

                    Text {
                        text: "AKMA GERÃ°LMESÃ° (İ,,â,€)"
                        color: "#6b7280"
                        font.family: "Segoe UI"
                        font.pixelSize: 10
                        font.letterSpacing: 1
                    }

                    Text {
                        id: tau0Metni
                        text: "â€""
                        color: "#3b82f6"
                        font.family: "Segoe UI"
                        font.pixelSize: 20
                        font.bold: true
                    }
                }
            }
        }

        Rectangle {

```

```

width: parent.width
height: 70
radius: 10
color: "#0a0e17"
border.color: "#241a38"
border.width: 1
clip: true

Rectangle { width: 4; height: parent.height; color: "#9333ea" }

Column {
    anchors.left: parent.left
    anchors.verticalCenter: parent.verticalCenter
    anchors.leftMargin: 20
    spacing: 2

    Text {
        text: "PLASTÄ°K VÄ°SKOZÄ°TE (İ¼)"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    Text {
        id: muMetni
        text: "â€""
        color: "#9333ea"
        font.family: "Segoe UI"
        font.pixelSize: 20
        font.bold: true
    }
}

Rectangle {
    width: parent.width
    height: 70
    radius: 10
    color: "#0a0e17"
    border.color: "#163321"
    border.width: 1
    clip: true

    Rectangle { width: 4; height: parent.height; color: "#16a34a" }

    Column {
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        anchors.leftMargin: 20
        spacing: 2

        Text {
            text: "UYUM KALÄ°TESÄ° (RÂ²)"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            id: r2Metni
            text: "â€""
            color: "#16a34a"
            font.family: "Segoe UI"
            font.pixelSize: 20
            font.bold: true
        }
    }
}

Rectangle {
    width: parent.width
    radius: 10
    color: "#0a0e17"

```

```

border.color: "#1e2a3f"
border.width: 1
height: tahminIcerik.implicitHeight + 24

Column {
    id: tahminIcerik
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.margins: 12
    spacing: 8

    Text {
        text: "BORU HATTI TAHMÄ°NÄ°"
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 10
        font.letterSpacing: 1
    }

    TextField {
        id: uzunlukKutusu
        width: parent.width
        height: 34
        placeholderText: "Boru UzunluÄŸu (m)"
        placeholderTextColor: "#4b5563"
        color: "ffffff"
        font.pixelSize: 12
        leftPadding: 10
        verticalAlignment: TextInput.AlignVCenter
        validator: DoubleValidator { bottom: 0; top: 200; decimals: 0 }
        background: Rectangle {
            color: "#060d17"
            radius: 6
            border.color: uzunlukKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
            border.width: 1
        }
    }

    TextField {
        id: debiKutusu
        width: parent.width
        height: 34
        placeholderText: "Hedef Debi (mÄ³/h)"
        placeholderTextColor: "#4b5563"
        color: "ffffff"
        font.pixelSize: 12
        leftPadding: 10
        verticalAlignment: TextInput.AlignVCenter
        validator: DoubleValidator { bottom: 0; top: 200; decimals: 1 }
        background: Rectangle {
            color: "#060d17"
            radius: 6
            border.color: debiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
            border.width: 1
        }
    }

    Button {
        width: parent.width
        height: 36
        text: "Tahmini Hesapla"
        font.pixelSize: 12
        font.bold: true

        background: Rectangle {
            radius: 6
            color: parent.hovered ? "#4f8cf7" : "#3b82f6"
            Behavior on color { ColorAnimation { duration: 120 } }
        }

        contentItem: Text {
            text: parent.text
            color: "ffffff"
            font: parent.font
        }
    }
}

```

```

        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }

    onClicked: {
        var L = parseFloat(uzunlukKutusu.text)
        var Q = parseFloat(debiKutusu.text)

        if (isNaN(L) || isNaN(Q) || L <= 0) {
            tahminSonucu.text = "GeÃ§erli deÃ§er girin"
            return
        }

        var basincMbar = (hesaplananTau0 + hesaplananMu * Q) * (L / 50)
        var basincBar = basincMbar / 1000
        tahminSonucu.text = basincBar.toFixed(2) + " bar"
    }
}

Text {
    id: tahminSonucu
    width: parent.width
    text: "â€"
    color: "#e8a020"
    font.family: "Segoe UI"
    font.pixelSize: 18
    font.bold: true
    horizontalAlignment: Text.AlignHCenter
}
}
}

Rectangle {
    id: durumKutusu
    width: parent.width
    height: durumIcerik.implicitHeight + 24
    radius: 10
    color: durumIyiMi ? "#0f2417" : "#2a1414"
    border.color: durumIyiMi ? "#16a34a" : "#dc2626"
    border.width: 1

    property bool durumIyiMi: true
    property bool tau0Yukse: false
    property bool muYukse: false

    Column {
        id: durumIcerik
        anchors.top: parent.top
        anchors.horizontalCenter: parent.horizontalCenter
        anchors.topMargin: 12
        spacing: 8
        width: parent.width - 24

        Row {
            anchors.horizontalCenter: parent.horizontalCenter
            spacing: 10

            Rectangle {
                width: 24
                height: 24
                radius: 12
                anchors.verticalCenter: parent.verticalCenter
                color: durumKutusu.durumIyiMi ? "#16a34a" : "#dc2626"

                Text {
                    anchors.centerIn: parent
                    text: durumKutusu.durumIyiMi ? "â€" : "â€—"
                    color: "#ffffff"
                    font.pixelSize: 13
                    font.bold: true
                }
            }

            Text {
                anchors.verticalCenter: parent.verticalCenter

```

```

        text: durumKutusu.durumIyiMi ? "POMPALANAB  L  R" : "POMPALAMA SORUNU"
        color: durumKutusu.durumIyiMi ? "#4ade80" : "#f87171"
        font.family: "Segoe UI"
        font.pixelSize: 15
        font.bold: true
    }
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    visible: !durumKutusu.durumIyiMi
    text: {
        if (durumKutusu.tau0Yuksek && durumKutusu.muYuksek) {
            return "Hem akma gerilmesi hem plastik viskozite y  ksek. Su/  şimento oran  n   art  r  n ve agrega gradasyonunu g  zden ge  şirin."
        } else if (durumKutusu.tau0Yuksek) {
            return "Akma gerilmesi y  ksek. Su/  şimento oran  n   art  r  may   veya s  perak    kanla  t  r  c   dozaj  n   y  kseltmeyi de  erlendirin."
        } else if (durumKutusu.muYuksek) {
            return "Plastik viskozite y  ksek.   nce agrega oran  n   azaltmay   veya su azalt  c   katkı eklemeyi de  erlendirin."
        }
        return "Yeterli stroke verisi bulunamad  ."
    }
    color: "#9ca3af"
    font.family: "Segoe UI"
    font.pixelSize: 11
    wrapMode: Text.WordWrap
}
}

Button {
    width: parent.width
    height: 40
    text: "    , PDF Rapor Olu  tur"
    font.pixelSize: 13
    font.bold: true
    enabled: olcumId > 0

    onClicked: {
        reportManager.pdfOlustur(
            olcumId,
            "Musteri",
            "Recete",
            hesaplananTau0,
            hesaplananMu,
            parseFloat(r2Metni.text)
        )
        console.log("PDF rapor olusturuldu.")
    }

    background: Rectangle {
        radius: 8
        color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
        Behavior on color { ColorAnimation { duration: 120 } }
    }

    contentItem: Text {
        text: parent.text
        color: "ffffff"
        font: parent.font
        horizontalAlignment: Text.AlignHCenter
        verticalAlignment: Text.AlignVCenter
    }
}

Button {
    width: parent.width
    height: 40
    text: "     Excel (CSV) Olarak D    ya Aktar"
    font.pixelSize: 13
    font.bold: true
    enabled: olcumId > 0

```



```

onClicked: {
    database.csvDisaAktar(olcumId)
    console.log("CSV disa aktarildi.")
}

background: Rectangle {
    radius: 8
    color: parent.enabled ? (parent.hovered ? "#22c55e" : "#16a34a") : "#1e2a3f"
    Behavior on color { ColorAnimation { duration: 120 } }
}

contentItem: Text {
    text: parent.text
    color: "#ffffff"
    font: parent.font
    horizontalAlignment: Text.AlignHCenter
    verticalAlignment: Text.AlignVCenter
}
}

Rectangle {
    width: parent.width
    radius: 10
    color: "#0a0e17"
    border.color: "#1e2a3f"
    border.width: 1
    height: playbackIcerik.implicitHeight + 24
    visible: olcumId > 0

    Column {
        id: playbackIcerik
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.margins: 12
        spacing: 8

        Text {
            text: "OYNATMA (PLAYBACK)"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 10
            font.letterSpacing: 1
        }

        Text {
            text: strokeModeli.count > 0
                ? "Stroke " + (playbackAdimi + 1) + " / " + strokeModeli.count
                : "Oynatılacak stroke yok"
            color: "#dce8f5"
            font.family: "Segoe UI"
            font.pixelSize: 12
        }
    }

    Row {
        width: parent.width
        spacing: 8

        Button {
            width: 44
            height: 34
            text: oynatiliyor ? "▶️" : "⏸️"
            font.pixelSize: 14
            enabled: strokeModeli.count > 0

            onClicked: {
                if (oynatiliyor) {
                    oynatiliyor = false
                    playbackTimer.stop()
                } else {
                    if (playbackAdimi >= strokeModeli.count - 1) {
                        playbackAdimi = -1
                    }
                    oynatiliyor = true
                }
            }
        }
    }
}

```

```

        playbackTimer.start()
    }
}

background: Rectangle {
    radius: 6
    color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
    Behavior on color { ColorAnimation { duration: 120 } }
}

contentItem: Text {
    text: parent.text
    color: "#ffffff"
    font: parent.font
    horizontalAlignment: Text.AlignHCenter
    verticalAlignment: Text.AlignVCenter
}
}

Repeater {
    model: [1, 2, 5, 10]

    Button {
        width: 44
        height: 34
        text: modelData + "x"
        font.pixelSize: 12

        onClicked: playbackHizi = modelData

        background: Rectangle {
            radius: 6
            color: playbackHizi === modelData ? "#3b82f6" : "#0f1420"
            border.color: "#1e2a3f"
            border.width: 1
        }

        contentItem: Text {
            text: parent.text
            color: playbackHizi === modelData ? "#ffffff" : "#9ca3af"
            font: parent.font
            horizontalAlignment: Text.AlignHCenter
            verticalAlignment: Text.AlignVCenter
        }
    }
}
}
}
}
}
}
}
}
}

Rectangle {
    id: panelAyraci
    width: 3
    height: parent.height
    color: "#05070b"

    Rectangle {
        anchors.right: parent.right
        anchors.top: parent.top
        anchors.bottom: parent.bottom
        width: 1
        color: "#403b82f6"
    }
}

Item {
    width: parent.width - ozetPaneli.width - panelAyraci.width
    height: parent.height

    Rectangle {
        id: grafikKutusu
        anchors.top: parent.top

```

```

anchors.left: parent.left
anchors.right: parent.right
anchors.margins: 16
height: parent.height * 0.55
radius: 10
color: "#0f1420"
border.color: "#1e2a3f"
border.width: 1

Item {
    id: grafikBaslik
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.margins: 14
    height: 20

    Text {
        anchors.left: parent.left
        anchors.verticalCenter: parent.verticalCenter
        text: "P-Q DaŸÄ±lÄ±m GrafiŸi"
        color: "#9ca3af"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

    Row {
        anchors.right: parent.right
        anchors.verticalCenter: parent.verticalCenter
        spacing: 16

        Row {
            spacing: 6
            anchors.verticalCenter: parent.verticalCenter

            Rectangle {
                width: 8
                height: 8
                radius: 4
                color: "#3b82f6"
                anchors.verticalCenter: parent.verticalCenter
            }

            Text {
                text: "Ä-lÄŸÄ±m NoktalarÄ±"
                color: "#6b7280"
                font.family: "Segoe UI"
                font.pixelSize: 11
                anchors.verticalCenter: parent.verticalCenter
            }
        }

        Row {
            spacing: 6
            anchors.verticalCenter: parent.verticalCenter

            Rectangle {
                width: 12
                height: 2
                color: "#e8a020"
                anchors.verticalCenter: parent.verticalCenter
            }

            Text {
                text: "Regresyon DoŸrusu"
                color: "#6b7280"
                font.family: "Segoe UI"
                font.pixelSize: 11
                anchors.verticalCenter: parent.verticalCenter
            }
        }

        Row {
            spacing: 6

```

```

        anchors.verticalCenter: parent.verticalCenter

        Rectangle {
            width: 8
            height: 8
            radius: 4
            color: "#e8a020"
            anchors.verticalCenter: parent.verticalCenter
        }

        Text {
            text: "Oynatma"
            color: "#6b7280"
            font.family: "Segoe UI"
            font.pixelSize: 11
            anchors.verticalCenter: parent.verticalCenter
        }
    }
}

```

```

ChartView {
    anchors.top: grafikBaslik.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.topMargin: 4
    anchors.margins: 8
    backgroundColor: "transparent"
    legend.visible: false
    antialiasing: true

    ValueAxis {
        id: qEkseni
        min: 0
        max: 20
        titleText: "Debi (Q) m3/h"
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ValueAxis {
        id: pEkseni
        min: 0
        max: 100
        titleText: "BasÄ±nÄŒ (P) mbar"
        gridLineColor: "#182131"
        labelsColor: "#4b5563"
        labelsFont.pixelSize: 9
        lineVisible: false
        minorGridVisible: false
    }

    ScatterSeries {
        id: pqSerisi
        axisX: qEkseni
        axisY: pEkseni
        color: "#3b82f6"
        markerSize: 10
    }

    LineSeries {
        id: regresyonCizgisi
        axisX: qEkseni
        axisY: pEkseni
        color: "#e8a020"
        width: 2
    }

    ScatterSeries {
        id: oynatmaSerisi
        axisX: qEkseni
    }
}

```

```

        axisY: pEksen1
        color: "#e8a020"
        markerSize: 16
    }
}
}

Rectangle {
    anchors.top: grafikKutusu.bottom
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.margins: 16
    anchors.topMargin: 8
    radius: 10
    color: "#0f1420"
    border.color: "#1e2a3f"
    border.width: 1

    Text {
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.margins: 14
        text: "Stroke Tablosu"
        color: "#9ca3af"
        font.family: "Segoe UI"
        font.pixelSize: 13
        font.bold: true
    }

    Row {
        id: tabloBasligi
        anchors.top: parent.top
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.topMargin: 36
        anchors.leftMargin: 12
        anchors.rightMargin: 12

        Text { width: parent.width * 0.25; text: "STROKE"; color: "#6b7280"; font.pixelSize: 11; font.bold: true;
font.letterSpacing: 1 }
        Text { width: parent.width * 0.25; text: "P (mbar)"; color: "#6b7280"; font.pixelSize: 11; font.bold:
true; font.letterSpacing: 1 }
        Text { width: parent.width * 0.25; text: "Q (mÅ³/h)"; color: "#6b7280"; font.pixelSize: 11; font.bold:
true; font.letterSpacing: 1 }
        Text { width: parent.width * 0.25; text: "DURUM"; color: "#6b7280"; font.pixelSize: 11; font.bold: true;
font.letterSpacing: 1 }
    }

    Rectangle {
        anchors.top: tabloBasligi.bottom
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.topMargin: 8
        anchors.leftMargin: 12
        anchors.rightMargin: 12
        height: 1
        color: "#1e2a3f"
    }

    ListView {
        anchors.top: tabloBasligi.bottom
        anchors.left: parent.left
        anchors.right: parent.right
        anchors.bottom: parent.bottom
        anchors.margins: 12
        anchors.topMargin: 14
        clip: true

        model: ListModel {
            id: strokeModel1
        }

        delegate: Rectangle {
            width: parent.width

```

```
height: 32
radius: 6
color: index === playbackAdimi ? "#1e2a3f" : (index % 2 === 0 ? "transparent" : "#0a0e17")

Row {
    anchors.fill: parent
    anchors.leftMargin: 6

    Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter; text: stroke;
color: "#dce8f5"; font.pixelSize: 12 }
    Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter; text:
p.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }
    Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter; text:
q.toFixed(2); color: "#dce8f5"; font.pixelSize: 12 }

    Item {
        width: parent.width * 0.25
        height: parent.height

        Rectangle {
            anchors.verticalCenter: parent.verticalCenter
            width: durumMetni.implicitWidth + 16
            height: 20
            radius: 10
            color: gecerli ? "#123321" : "#2a1414"

            Text {
                id: durumMetni
                anchors.centerIn: parent
                text: gecerli ? "GeÃ§erli" : "HatalÃ±"
                color: gecerli ? "#4ade80" : "#f87171"
                font.pixelSize: 11
                font.bold: true
            }
        }
    }
}
}
```

src\Backend.cpp

```
#include "Backend.h"
#include <QDebug>

Backend::Backend(QObject *parent)
    : QObject(parent)
{
}

QString Backend::durum() const
{
    return "Baglanti bekleniyor...";
}

bool Backend::girisYap(const QString &kullaniciAdi, const QString &sifre)
{
    if (kullaniciAdi == "admin" && sifre == "admin123") {
        qDebug() << "Giris Basarili:" << kullaniciAdi;
        return true;
    }

    qDebug() << "Giris Basarisiz:" << kullaniciAdi;
    return false;
}
```

src\Backend.h

```
#pragma once
#include <QObject>

class Backend : public QObject
{
    Q_OBJECT
    Q_PROPERTY(QString durum READ durum CONSTANT)

public:
    explicit Backend(QObject *parent = nullptr);
    QString durum() const;
    Q_INVOKABLE bool girisYap(const QString &kullaniciAdi, const QString &sifre);
};
```


src\Calculator.cpp

```
#include "Calculator.h"

Calculator::Calculator(QObject *parent)
    : QObject(parent)
{
}

QString Calculator::durum() const
{
    return m_durum;
}

int Calculator::strokeSayisi() const
{
    return m_strokeSayisi;
}

bool Calculator::duraklatildi() const
{
    return m_duraklatildi;
}

void Calculator::konumGuncelle(double konum)
{
    if (m_duraklatildi) {
        return;
    }

    QString yeniDurum;

    if (konum > 450) {
        yeniDurum = "YUKARIDA";
    } else if (konum > 20) {
        yeniDurum = "INIYOR";
    } else {
        yeniDurum = "TAMAMLANDI";
    }

    if (yeniDurum == "TAMAMLANDI" && m_durum != "TAMAMLANDI") {
        m_strokeSayisi++;
        emit strokeSayisiChanged();
    }

    if (yeniDurum != m_durum) {
        m_durum = yeniDurum;
        emit durumChanged();
    }
}

void Calculator::duraklat()
{
    if (m_duraklatildi) {
        return;
    }

    m_duraklatildi = true;
    emit duraklatildiChanged();
}

void Calculator::devamEt()
{
    if (!m_duraklatildi) {
        return;
    }

    m_duraklatildi = false;
    emit duraklatildiChanged();
}

void Calculator::sifirla()
{
    m_strokeSayisi = 0;
    m_durum = "BEKLENIYOR";
}
```

```
m_duraklatildi = false;  
emit strokeSayisiChanged();  
emit durumChanged();  
emit duraklatildiChanged();  
}
```

src\Calculator.h

```
#pragma once
#include <QObject>

class Calculator : public QObject
{
    Q_OBJECT
    Q_PROPERTY(QString durum READ durum NOTIFY durumChanged)
    Q_PROPERTY(int strokeSayisi READ strokeSayisi NOTIFY strokeSayisiChanged)
    Q_PROPERTY(bool duraklatildi READ duraklatildi NOTIFY duraklatildiChanged)

public:
    explicit Calculator(QObject *parent = nullptr);
    QString durum() const;
    int strokeSayisi() const;
    bool duraklatildi() const;

    Q_INVOKABLE void konumGuncelle(double konum);
    Q_INVOKABLE void duraklat();
    Q_INVOKABLE void devamEt();
    Q_INVOKABLE void sifirla();

signals:
    void durumChanged();
    void strokeSayisiChanged();
    void duraklatildiChanged();

private:
    QString m_durum = "BEKLENIYOR";
    int m_strokeSayisi = 0;
    bool m_duraklatildi = false;
};
```

src\Database.cpp

```

#include "Database.h"
#include <QSqlQuery>
#include <QSqlError>
#include <QDebug>
#include <QDateTime>
#include <QVector>
#include <QFile>
#include <QTextStream>
#include <QStandardPaths>
#include <QDir>

Database::Database(QObject *parent)
    : QObject(parent)
{
    m_db = QSqlDatabase::addDatabase("QSQLITE");
    m_db.setDatabaseName("sliper.db");

    if (!m_db.open()) {
        qWarning() << "Veritabani acilamadi:" << m_db.lastError().text();
    } else {
        qDebug() << "Veritabani basariyla acildi.";
        tablolariOlustur();
    }
}

void Database::tablolariOlustur()
{
    QSqlQuery sorgu;

    sorgu.exec(
        "CREATE TABLE IF NOT EXISTS Olcumler ("
        "id INTEGER PRIMARY KEY AUTOINCREMENT, "
        "tarih TEXT, "
        "musteri TEXT, "
        "recete TEXT, "
        "agirlik REAL"
        ")"
    );

    sorgu.exec(
        "CREATE TABLE IF NOT EXISTS Strokelar ("
        "id INTEGER PRIMARY KEY AUTOINCREMENT, "
        "olcumId INTEGER, "
        "basinc REAL, "
        "konum REAL, "
        "debi REAL, "
        "gecerli INTEGER, "
        "FOREIGN KEY(olcumId) REFERENCES Olcumler(id)"
        ")"
    );

    QSqlQuery migrasyon;
    migrasyon.exec("ALTER TABLE Strokelar ADD COLUMN debi REAL DEFAULT 0");

    sorgu.exec(
        "CREATE TABLE IF NOT EXISTS Kalibrasyonlar ("
        "sensor TEXT PRIMARY KEY, "
        "deger1 REAL, "
        "deger2 REAL, "
        "tarih TEXT"
        ")"
    );

    qDebug() << "Tablolar hazir.";
}

int Database::olcumBaslat(const QString &musteri, const QString &recete, double agirlik)
{
    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO Olcumler (tarih, muster, recete, agirlik) "
        "VALUES (:tarih, :musteri, :recete, :agirlik)"
    );
}

```

```

    sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
    sorgu.bindValue(":musteri", musteri);
    sorgu.bindValue(":recete", recete);
    sorgu.bindValue(":agirlik", agirlik);

    if (!sorgu.exec()) {
        qWarning() << "Olcum baslatilamadi:" << sorgu.lastError().text();
        return -1;
    }

    int yeniId = sorgu.lastInsertId().toInt();
    qDebug() << "Yeni olcum baslatildi, id:" << yeniId;
    return yeniId;
}

void Database::strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli)
{
    if (olcumId <= 0) {
        qWarning() << "Gecersiz olcumId, stroke kaydedilemedi.";
        return;
    }

    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO Strokelar (olcumId, basinc, konum, debi, gecerli) "
        "VALUES (:olcumId, :basinc, :konum, :debi, :gecerli)"
    );

    sorgu.bindValue(":olcumId", olcumId);
    sorgu.bindValue(":basinc", basinc);
    sorgu.bindValue(":konum", konum);
    sorgu.bindValue(":debi", debi);
    sorgu.bindValue(":gecerli", gecerli ? 1 : 0);

    if (!sorgu.exec()) {
        qWarning() << "Stroke kaydedilemedi:" << sorgu.lastError().text();
    } else {
        qDebug() << "Stroke kaydedildi, olcumId:" << olcumId;
    }
}

QVariantList Database::tumOlcumleriGetir()
{
    QVariantList sonuclar;

    QSqlQuery sorgu(
        "SELECT o.id, o.tarih, o.musteri, o.recete, o.agirlik, "
        "COUNT(s.id) AS strokeSayisi "
        "FROM Olcumler o "
        "LEFT JOIN Strokelar s ON s.olcumId = o.id "
        "GROUP BY o.id "
        "ORDER BY o.id DESC"
    );

    while (sorgu.next()) {
        QVariantMap satir;
        satir["id"] = sorgu.value("id");
        satir["tarih"] = sorgu.value("tarih");
        satir["musteri"] = sorgu.value("musteri");
        satir["recete"] = sorgu.value("recete");
        satir["agirlik"] = sorgu.value("agirlik");
        satir["strokeSayisi"] = sorgu.value("strokeSayisi");
        sonuclar.append(satir);
    }

    return sonuclar;
}

QVariantList Database::strokeVerileriGetir(int olcumId)
{
    QVariantList sonuclar;

    QSqlQuery sorgu;
    sorgu.prepare(

```

```

        "SELECT basinc, konum, debi, gecerli FROM Strokelar "
        "WHERE olcumId = :olcumId ORDER BY id ASC"
    );
    sorgu.bindValue(":olcumId", olcumId);

    if (!sorgu.exec()) {
        qWarning() << "Stroke verileri alinamadi:" << sorgu.lastError().text();
        return sonuclar;
    }

    int sira = 1;
    while (sorgu.next()) {
        QVariantMap satir;
        satir["stroke"] = sira++;
        satir["basinc"] = sorgu.value("basinc");
        satir["konum"] = sorgu.value("konum");
        satir["debi"] = sorgu.value("debi");
        satir["gecerli"] = sorgu.value("gecerli").toBool();
        sonuclar.append(satir);
    }

    return sonuclar;
}

QVariantMap Database::binghamHesapla(int olcumId)
{
    QVariantMap sonuc;

    QSqlQuery sorgu;
    sorgu.prepare(
        "SELECT basinc, debi FROM Strokelar "
        "WHERE olcumId = :olcumId AND gecerli = 1"
    );
    sorgu.bindValue(":olcumId", olcumId);

    QVector<double> pDegerleri;
    QVector<double> qDegerleri;

    if (sorgu.exec()) {
        while (sorgu.next()) {
            pDegerleri.append(sorgu.value("basinc").toDouble());
            qDegerleri.append(sorgu.value("debi").toDouble());
        }
    }

    const int n = pDegerleri.size();

    if (n < 2) {
        sonuc["tau0"] = 0.0;
        sonuc["mu"] = 0.0;
        sonuc["r2"] = 0.0;
        sonuc["yeterliVeri"] = false;
        return sonuc;
    }

    double qToplam = 0, pToplam = 0, qpToplam = 0, qKareToplam = 0;
    for (int i = 0; i < n; ++i) {
        qToplam += qDegerleri[i];
        pToplam += pDegerleri[i];
        qpToplam += qDegerleri[i] * pDegerleri[i];
        qKareToplam += qDegerleri[i] * qDegerleri[i];
    }

    const double qOrt = qToplam / n;
    const double pOrt = pToplam / n;

    const double payda = qKareToplam - n * qOrt * qOrt;
    const double mu = (payda != 0.0) ? (qpToplam - n * qOrt * pOrt) / payda : 0.0;
    const double tau0 = pOrt - mu * qOrt;

    double ssTot = 0, ssRes = 0;
    for (int i = 0; i < n; ++i) {
        const double tahmin = tau0 + mu * qDegerleri[i];
        ssRes += (pDegerleri[i] - tahmin) * (pDegerleri[i] - tahmin);
        ssTot += (pDegerleri[i] - pOrt) * (pDegerleri[i] - pOrt);
    }

```

```

    }

    const double r2 = (ssTot != 0.0) ? (1.0 - ssRes / ssTot) : 0.0;

    sonuc["tau0"] = tau0;
    sonuc["mu"] = mu;
    sonuc["r2"] = r2;
    sonuc["yeterliVeri"] = true;
    return sonuc;
}

void Database::kalibrasyonKaydet(const QString &sensor, double deger1, double deger2)
{
    QSqlQuery sorgu;
    sorgu.prepare(
        "INSERT INTO Kalibrasyonlar (sensor, deger1, deger2, tarih) "
        "VALUES (:sensor, :deger1, :deger2, :tarih) "
        "ON CONFLICT(sensor) DO UPDATE SET "
        "deger1 = :deger1, deger2 = :deger2, tarih = :tarih"
    );

    sorgu.bindValue(":sensor", sensor);
    sorgu.bindValue(":deger1", deger1);
    sorgu.bindValue(":deger2", deger2);
    sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));

    if (!sorgu.exec()) {
        qWarning() << "Kalibrasyon kaydedilemedi:" << sorgu.lastError().text();
    } else {
        qDebug() << "Kalibrasyon kaydedildi:" << sensor;
    }
}

QVariantMap Database::kalibrasyonGetir(const QString &sensor)
{
    QVariantMap sonuc;

    QSqlQuery sorgu;
    sorgu.prepare("SELECT deger1, deger2, tarih FROM Kalibrasyonlar WHERE sensor = :sensor");
    sorgu.bindValue(":sensor", sensor);

    if (sorgu.exec() && sorgu.next()) {
        sonuc["deger1"] = sorgu.value("deger1");
        sonuc["deger2"] = sorgu.value("deger2");
        sonuc["tarih"] = sorgu.value("tarih");
        sonuc["mevcut"] = true;
    } else {
        sonuc["mevcut"] = false;
    }

    return sonuc;
}

QVariantMap Database::olcumBilgisiGetir(int olcumId)
{
    QVariantMap sonuc;

    QSqlQuery sorgu;
    sorgu.prepare("SELECT tarih, musteri, recete, agirlik FROM Olcumler WHERE id = :id");
    sorgu.bindValue(":id", olcumId);

    if (sorgu.exec() && sorgu.next()) {
        sonuc["tarih"] = sorgu.value("tarih");
        sonuc["musteri"] = sorgu.value("musteri");
        sonuc["recete"] = sorgu.value("recete");
        sonuc["agirlik"] = sorgu.value("agirlik");
        sonuc["bulundu"] = true;
    } else {
        sonuc["bulundu"] = false;
    }

    return sonuc;
}

bool Database::csvDisaAktar(int olcumId)

```

```

{
    QVariantMap bilgi = olcumBilgisiGetir(olcumId);
    if (!bilgi["bulundu"].toBool()) {
        qWarning() << "Olcum bulunamadi, csv olusturulamadi.";
        return false;
    }

    QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) + "/SliperRaporlari";
    QDir().mkpath(klasor);

    QString dosyaAdi = klasor + QString("/SLIPER_Veri_%1_%2.csv")
        .arg(olcumId)
        .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));

    QFile dosya(dosyaAdi);
    if (!dosya.open(QIODevice::WriteOnly | QIODevice::Text)) {
        qWarning() << "CSV dosyasi acilamadi:" << dosyaAdi;
        return false;
    }

    QTextStream akis(&dosya);
    akis.setEncoding(QStringConverter::Utf8);

    akis << "Olcum ID;" << olcumId << "\n";
    akis << "Tarih;" << bilgi["tarih"].toString() << "\n";
    akis << "Musteri;" << bilgi["musteri"].toString() << "\n";
    akis << "Recete;" << bilgi["recete"].toString() << "\n";
    akis << "Agirlik (kg);" << bilgi["agirlik"].toDouble() << "\n";
    akis << "\n";
    akis << "Stroke;Basinc (mbar);Konum (mm);Debi (m3/h);Gecerli\n";

    QVariantList strokelar = strokeVerileriGetir(olcumId);
    for (const QVariant &v : strokelar) {
        QVariantMap s = v.toMap();
        akis << s["stroke"].toInt() << ";";
        akis << s["basinc"].toDouble() << ";";
        akis << s["konum"].toDouble() << ";";
        akis << s["debi"].toDouble() << ";";
        akis << (s["gecerli"].toBool() ? "Evet" : "Hayir") << "\n";
    }

    dosya.close();
    qDebug() << "CSV disa aktarildi:" << dosyaAdi;
    return true;
}

```


src\Database.h

```
#pragma once
#include <QObject>
#include <QSqlDatabase>
#include <QVariantList>
#include <QVariantMap>

class Database : public QObject
{
    Q_OBJECT

public:
    explicit Database(QObject *parent = nullptr);

    Q_INVOKABLE int olcumBaslat(const QString &musteri, const QString &recete, double agirlik);
    Q_INVOKABLE void strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli);
    Q_INVOKABLE QVariantList tumOlcumleriGetir();
    Q_INVOKABLE QVariantList strokeVerileriGetir(int olcumId);
    Q_INVOKABLE QVariantMap binghamHesapla(int olcumId);
    Q_INVOKABLE void kalibrasyonKaydet(const QString &sensor, double deger1, double deger2);
    Q_INVOKABLE QVariantMap kalibrasyonGetir(const QString &sensor);
    Q_INVOKABLE QVariantMap olcumBilgisiGetir(int olcumId);
    Q_INVOKABLE bool csvDisaAktar(int olcumId);

private:
    void tablolariOlustur();
    QSqlDatabase m_db;
};
```

src\ReportManager.cpp

```

#include "ReportManager.h"
#include <QPrinter>
#include <QPainter>
#include <QTextDocument>
#include <QStandardPaths>
#include <QDir>
#include <QDateTime>
#include <QDebug>

ReportManager::ReportManager(QObject *parent)
    : QObject(parent)
{
}

bool ReportManager::pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
                                double tau0, double mu, double r2)
{
    QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) + "/SliperRaporlari";
    QDir().mkpath(klasor);

    QString dosyaAdi = klasor + QString("/SLIPER_Rapor_%1_%2.pdf")
        .arg(olcumId)
        .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));

    QTextDocument belge;
    QString html = QString(
        "<h1 style='color:#3b82f6;'>SLIPER Analiz Raporu</h1>"
        "<p><b>Olcum ID:</b> %1</p>"
        "<p><b>Musteri:</b> %2</p>"
        "<p><b>Recete:</b> %3</p>"
        "<p><b>Tarih:</b> %4</p>"
        "<hr>"
        "<h2>Bingham Model Sonuclari</h2>"
        "<p><b>Akma Gerilmesi (tau0):</b> %5 mbar</p>"
        "<p><b>Plastik Viskozite (mu):</b> %6 mbar*h/m3</p>"
        "<p><b>Uyum Kalitesi (R2):</b> %7</p>"
        "<hr>"
        "<p style='color:#6b7280; font-size:10px;'>Liya Laboratuvar Test Cihazlari - SLIPER Analiz Sistemi</p>"
    ).arg(olcumId)
        .arg(musteri)
        .arg(recete)
        .arg(QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"))
        .arg(QString::number(tau0, 'f', 2))
        .arg(QString::number(mu, 'f', 2))
        .arg(QString::number(r2, 'f', 2));

    belge.setHtml(html);

    QPrinter yazici(QPrinter::HighResolution);
    yazici.setOutputFormat(QPrinter::PdfFormat);
    yazici.setOutputFileName(dosyaAdi);

    belge.print(&yazici);

    qDebug() << "PDF rapor olusturuldu:" << dosyaAdi;
    return true;
}

```

src\ReportManager.h

```
#pragma once
#include <QObject>

class ReportManager : public QObject
{
    Q_OBJECT

public:
    explicit ReportManager(QObject *parent = nullptr);
    Q_INVOKABLE bool pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
                                double tau0, double mu, double r2);
};
```

src\SensorManager.cpp

```
#include "SensorManager.h"

SensorManager::SensorManager(QObject *parent)
    : QObject(parent)
{
}

double SensorManager::basinc() const
{
    return m_basinc;
}

double SensorManager::konum() const
{
    return m_konum;
}

double SensorManager::hiz() const
{
    return m_hiz;
}

double SensorManager::debi() const
{
    return m_debi;
}

bool SensorManager::veriGecerli() const
{
    return m_veriGecerli;
}

void SensorManager::veriGuncelle(double basinc, double konum, double hiz, double debi)
{
    const bool veriGecerliDegisti = !m_veriGecerli;
    m_veriGecerli = true;

    if (m_basinc != basinc) {
        m_basinc = basinc;
        emit basincChanged();
    }

    if (m_konum != konum) {
        m_konum = konum;
        emit konumChanged();
    }

    if (m_hiz != hiz) {
        m_hiz = hiz;
        emit hizChanged();
    }

    if (m_debi != debi) {
        m_debi = debi;
        emit debiChanged();
    }

    if (veriGecerliDegisti) {
        emit veriGecerliChanged();
    }
}

void SensorManager::veriyiGecersizYap()
{
    if (!m_veriGecerli) {
        return;
    }

    m_veriGecerli = false;
    emit veriGecerliChanged();
}
```

src\SensorManager.h

```
#pragma once
#include <QObject>

class SensorManager : public QObject
{
    Q_OBJECT
    Q_PROPERTY(double basinc READ basinc NOTIFY basincChanged)
    Q_PROPERTY(double konum READ konum NOTIFY konumChanged)
    Q_PROPERTY(double hiz READ hiz NOTIFY hizChanged)
    Q_PROPERTY(double debi READ debi NOTIFY debiChanged)
    Q_PROPERTY(bool veriGecerli READ veriGecerli NOTIFY veriGecerliChanged)

public:
    explicit SensorManager(QObject *parent = nullptr);
    double basinc() const;
    double konum() const;
    double hiz() const;
    double debi() const;
    bool veriGecerli() const;

    Q_INVOKABLE void veriGuncelle(double basinc, double konum, double hiz, double debi);
    Q_INVOKABLE void veriyiGecersizYap();

signals:
    void basincChanged();
    void konumChanged();
    void hizChanged();
    void debiChanged();
    void veriGecerliChanged();

private:
    double m_basinc = 0.0;
    double m_konum = 0.0;
    double m_hiz = 0.0;
    double m_debi = 0.0;
    bool m_veriGecerli = false;
};
```